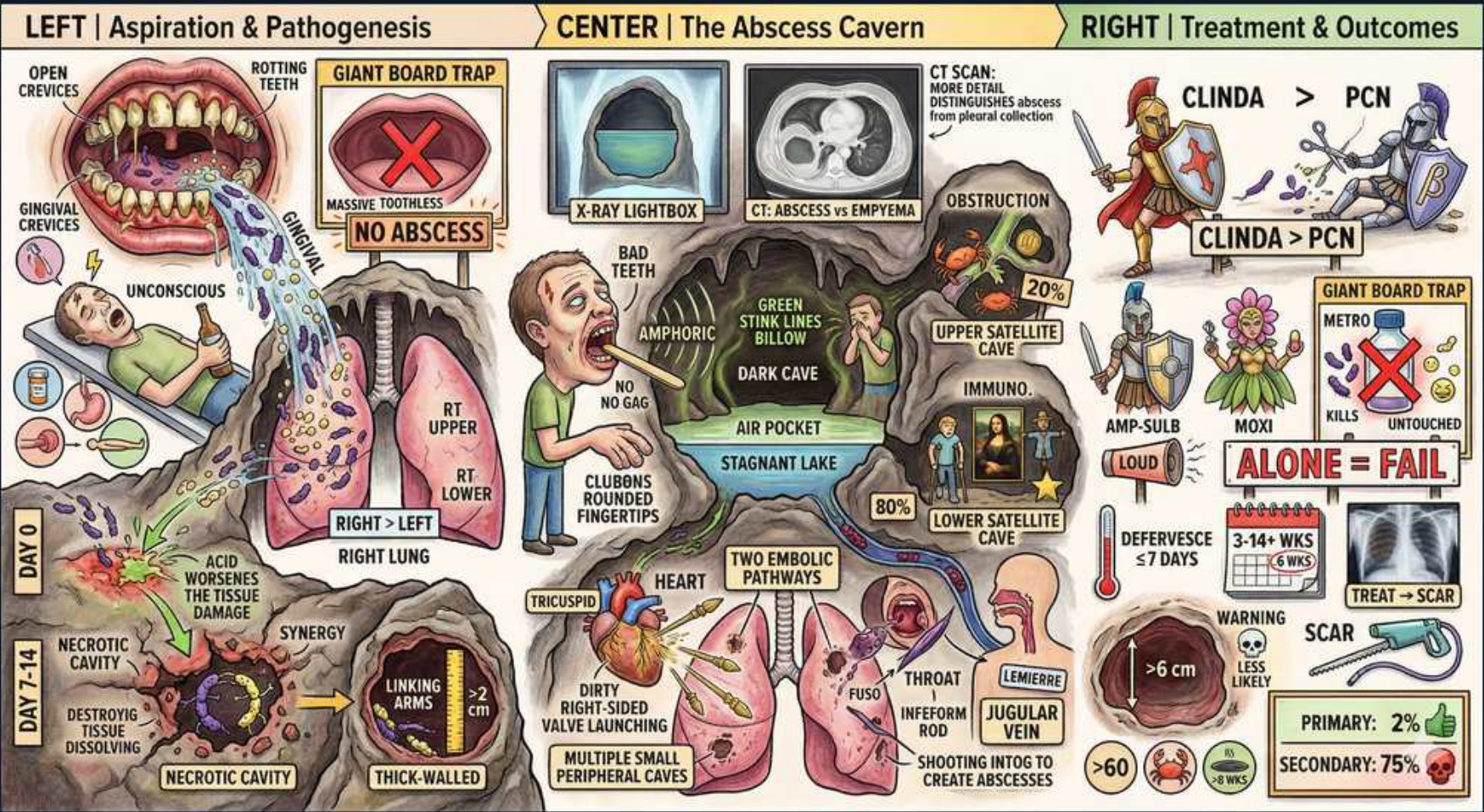


Lung Abscess — Pathogenesis, Imaging & Treatment



1. Poor dentition / periodontal source

WHAT YOU SEE

Rotting teeth, gingival crevices

WHAT IT ENCODES

Poor oral hygiene and periodontal disease provide the anaerobic bacterial inoculum for aspiration lung abscess. More teeth/plaque means higher anaerobic burden. Edentulous patients rarely form anaerobic abscess (suspect obstruction/cancer instead).

BOARD TRAP

An edentulous patient with a 'lung abscess' should raise suspicion for an obstructing tumor — anaerobic abscess needs a dental source.

2. Aspiration / unconscious

WHAT YOU SEE

Unconscious patient supine

WHAT IT ENCODES

Risk factors are impaired consciousness and impaired swallow: alcohol, seizures, stroke, anesthesia, dysphagia. Aspirated oropharyngeal anaerobes seed dependent lung segments.

BOARD TRAP

Loss of consciousness (alcohol/seizure) is the classic setup — primary abscess is the result of aspiration, not hematogenous spread.

3. Dependent segments (RUL post, RLL sup)

WHAT YOU SEE

Right lung 'RT UPPER / RT LOWER' marked

WHAT IT ENCODES

Aspiration in the supine position seeds gravity-dependent segments: posterior segment of the right upper lobe and superior segment of the right lower lobe. The right bronchus is more vertical, so the right lung is favored.

BOARD TRAP

Right-sided, dependent-segment location is classic for aspiration abscess — left-sided or atypical location should prompt other diagnoses.

4. Anaerobes / polymicrobial

WHAT YOU SEE

Necrotic cavity, tissue dissolving, synergy

WHAT IT ENCODES

Primary lung abscess is typically polymicrobial anaerobic (*Peptostreptococcus*, *Prevotella*, *Bacteroides*, *Fusobacterium*) with synergy. Tissue necrosis produces the cavity over 7–14 days.

BOARD TRAP

Anaerobic abscess produces foul/putrid sputum; cultures are often unhelpful because of contamination — treat empirically for anaerobes.

5. Putrid sputum / no abscess clue

WHAT YOU SEE

'GIANT BOARD TRAP: massive toothless = NO ABSCESS'

WHAT IT ENCODES

Foul-smelling (putrid) sputum strongly suggests anaerobic lung abscess. Conversely, a toothless/edentulous patient lacks the anaerobic source, so abscess is unlikely — look for obstruction or another cause.

BOARD TRAP

Edentulous + cavity = think malignancy/obstruction, NOT anaerobic abscess. The board trap is assuming all cavities are infectious.

6. CXR — air-fluid level

WHAT YOU SEE

X-ray lightbox, cavity

WHAT IT ENCODES

Chest X-ray shows a thick-walled cavity with an air-fluid level. A round opacity that cavitates over days is characteristic. Air-fluid level indicates communication with the airway.

BOARD TRAP

An air-fluid level on CXR indicates a cavity communicating with bronchus (abscess) — distinguish from a fluid-filled empyema, which needs drainage not just antibiotics.

7. CT — abscess vs empyema

WHAT YOU SEE

CT scan: 'abscess vs empyema', pleural collection

WHAT IT ENCODES

CT differentiates lung abscess (round, thick irregular wall, acute angle with pleura) from empyema (lenticular, smooth wall, 'split pleura' sign). This distinction changes management: drain empyema, usually don't drain abscess.

BOARD TRAP

Empyema needs chest-tube drainage; an uncomplicated abscess usually drains spontaneously via the airway and is managed with antibiotics alone — the CT split-pleura sign separates them.

8. Amphoric breath / clubbing

WHAT YOU SEE

Amphoric sound, clubbed fingertips

WHAT IT ENCODES

Examination may reveal amphoric/cavernous breath sounds over the cavity and digital clubbing with chronicity. Chronic suppuration drives clubbing.

BOARD TRAP

Clubbing implies chronicity (weeks); an acute presentation without clubbing does not exclude abscess but suggests a shorter course.

9. Clindamycin > penicillin

WHAT YOU SEE

Knight 'CLINDA > PCN'

WHAT IT ENCODES

Historically clindamycin outperformed penicillin for anaerobic lung abscess due to beta-lactamase–producing anaerobes; ampicillin-sulbactam or amoxicillin-clavulanate are now first-line. Metronidazole alone fails (misses microaerophilic streptococci).

BOARD TRAP

Metronidazole monotherapy fails lung abscess — it doesn't cover aerobic/microaerophilic streptococci; always combine or use a beta-lactam/beta-lactamase inhibitor.

10. Ampicillin-sulbactam / moxifloxacin

WHAT YOU SEE

'AMP-SULB', 'MOXI' options

WHAT IT ENCODES

Preferred regimens: ampicillin-sulbactam, amoxicillin-clavulanate, or moxifloxacin. These cover the polymicrobial anaerobic-aerobic mix. Carbapenems for resistant/complicated cases.

BOARD TRAP

Choose a beta-lactamase inhibitor combination or moxifloxacin — a plain beta-lactam may be inactivated by beta-lactamase–producing anaerobes.

11. Prolonged duration / defervesce <7d

WHAT YOU SEE

'DEFERVESCE <7 DAYS', '3-14+ wks'

WHAT IT ENCODES

Therapy is prolonged — typically several weeks until cavity closure (often 3–8+ weeks), guided by radiographic resolution. Fever should resolve within ~7 days; persistent fever suggests inadequate drainage or wrong diagnosis.

BOARD TRAP

Stopping antibiotics when the patient feels better (a few days) causes relapse — treat for weeks until imaging shows resolution.

12. Drainage if >6 cm / no response

WHAT YOU SEE

'>6 cm', 'TREAT → SCAR', warning

WHAT IT ENCODES

Most abscesses drain via the bronchus and respond to antibiotics. Percutaneous or bronchoscopic drainage is reserved for failure of medical therapy, very large cavities (>6 cm), or sepsis. Surgery is rarely needed.

BOARD TRAP

Routine drainage is NOT first-line for primary lung abscess — most resolve with antibiotics; intervene only for failure, large size, or hemorrhage.

13. Right-sided endocarditis source

WHAT YOU SEE

Tricuspid heart, 'dirty right-sided valve'

WHAT IT ENCODES

Septic emboli from right-sided (tricuspid) infective endocarditis — classically in IV drug users with *S. aureus* — produce multiple peripheral cavitory lung lesions, distinct from primary aspiration abscess.

BOARD TRAP

Multiple bilateral peripheral cavities suggest septic emboli from tricuspid endocarditis (IVDU), not single aspiration abscess — get blood cultures and echo.

14. Lemierre / jugular vein septic emboli

WHAT YOU SEE

'LEMIERRE', jugular vein, *Fusobacterium*

WHAT IT ENCODES

Lemierre syndrome: *Fusobacterium necrophorum* oropharyngeal infection causes septic internal jugular vein thrombophlebitis, showering septic emboli to the lungs. Suspect with recent pharyngitis and neck tenderness.

BOARD TRAP

Lemierre (*Fusobacterium*, jugular thrombophlebitis) is the trap for septic pulmonary emboli after a sore throat — not simple aspiration.

15. Primary vs secondary mortality

WHAT YOU SEE

'PRIMARY 2%, SECONDARY 75%'

WHAT IT ENCODES

Primary (aspiration) abscess has low mortality (~2%) and responds well to antibiotics. Secondary abscess (obstruction, immunocompromise, resistant organisms) carries far higher mortality (up to ~75%).

BOARD TRAP

The key prognostic split is primary vs secondary — secondary abscess (tumor/immunosuppression) is the deadly one and demands a search for the underlying cause.